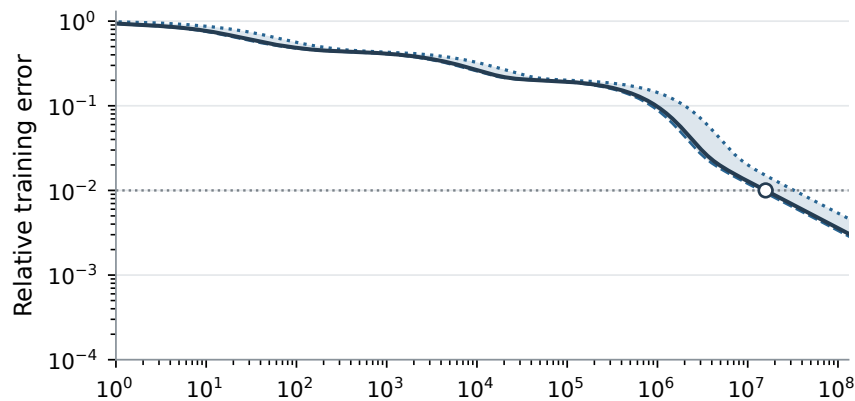
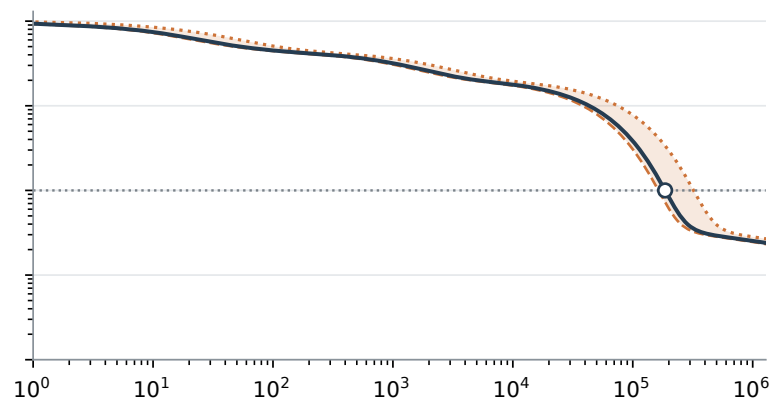


Theorem rate caps predict a lower bound on output error

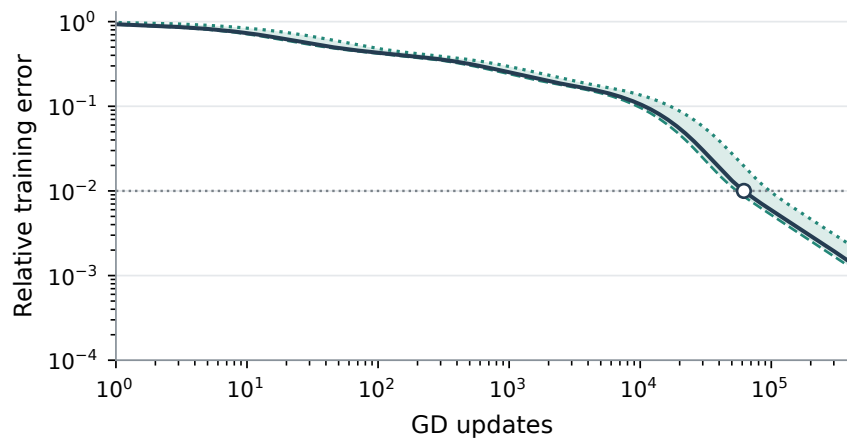
$\gamma = 8$ $13,934,809 \leq \text{steps} \leq 32,778,524$



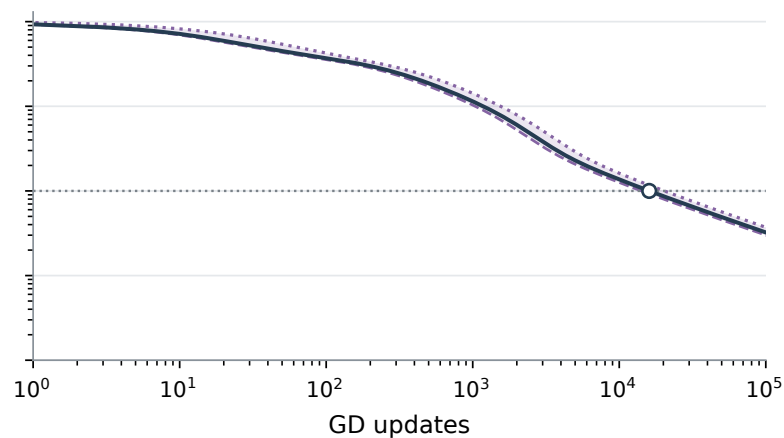
$\gamma = 12$ $163,954 \leq \text{steps} \leq 321,441$



$\gamma = 16$ $54,453 \leq \text{steps} \leq 97,205$



$\gamma = 64$ $14,118 \leq \text{steps} \leq 20,278$



--- Theorem output-error lower bound
... Companion output-error upper bound

— Finite-spectrum error
○ Executed GD 1% crossing